EXCEL CONDITIONAL FORMATTING

Complete Learning Handbook | Lessons 28-36

Concepts • Exact Excel steps • Sample data • Expected results • Custom rules • Pivot Table integration

Scope: all nine lessons from the course transcript and our worked examples. Practice assignments have been excluded.

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Lesson	Topic
28	Conditional Formatting fundamentals
29	Data Bars: solid, gradient, show bar only
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33	Icon Sets: arrows, ratings, thresholds, target references
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36	Integrated conditional formatting and Slicer reporting

Course conventions

All instructions use Microsoft Excel desktop ribbon names. Some labels vary slightly across Excel versions. For examples, use workbook Excel_Complete_Practice.xlsx; create a new sheet for each lesson. When source values are shown as percentages, enter them as real Excel percentages (for instance, 8% is stored as 0.08).

Important: Conditional Formatting changes how cells appear, not their underlying values. Normal source-data changes update formatting automatically; Pivot Tables may require Refresh to reflect source changes.

LESSON 28 | Conditional Formatting Fundamentals

Definition and purpose

Conditional Formatting automatically applies cell formatting according to a rule. Use it to identify high/low values, duplicates, outliers, targets, or trends at a glance. Pivot Tables summarize figures; Conditional Formatting makes patterns visible.

Where to find the tools

Excel menu	Purpose
Home > Conditional Formatting > Highlight Cells Rules	Greater Than, Less Than, Between, Equal To, Text, Date, Duplicate Values
Top/Bottom Rules	Top/bottom items or percentages, above/below average
Data Bars	In-cell bars for magnitude comparisons
Color Scales	Two-/three-color value gradients
Icon Sets	Status and rating symbols
New Rule / Manage Rules / Clear Rules	Create, edit, inspect, prioritize or remove rules

Practice dataset: 28_ConditionalFormatting

Date	Sales
01-Nov	6500
02-Nov	8200
03-Nov	10842
04-Nov	7100
05-Nov	9300
06-Nov	5800
07-Nov	4943
08-Nov	7800
09-Nov	8900
10-Nov	6200

Example A: Greater Than 8000

1. Select B2:B11 (numbers only).

2. Home > Conditional Formatting > Highlight Cells Rules > Greater Than.

3. Enter 8000, choose a green fill, click OK.

Expected: 8200, 10842, 9300, and 8900 highlighted (four cells).

Example B: Less Than 6000

1. Keep B2:B11 selected.

2. Home > Conditional Formatting > Highlight Cells Rules > Less Than.

3. Enter 6000, choose a red fill, click OK.

Expected: 5800 and 4943 highlighted; both rules coexist on the same range.

Dynamic updates and rule management

Change B2 from 6500 to 9500: it becomes green automatically; restore 6500 and formatting disappears. Open Home > Conditional Formatting > Manage Rules to edit conditions, formats, Applies To ranges, rule order, and Stop If True. Use Clear Rules > Clear Rules from Selected Cells to remove formatting but keep data.

LESSON 29 | Data Bars

What they do

Data Bars draw horizontal in-cell bars whose lengths reflect numeric magnitude. Solid Fill has a uniform color; Gradient Fill fades across each bar. Both use the same numeric scaling.

Practice dataset: 29_DataBars

Product	Sales
Crispy Pizza	30000
Cheese Pizza	25000
Paneer Tikka Pizza	45000
Crispy Burger	35000
Veg Burger	15000
Aloo Shorts	10000

Apply solid or gradient bars

1. Select B2:B7.

2. Home > Conditional Formatting > Data Bars.

3. Choose a Solid Fill preset; hover for Live Preview. To compare, clear the existing rule and apply a Gradient Fill preset.

Expected order: Paneer Tikka Pizza longest, then Crispy Burger, Crispy Pizza, Cheese Pizza, Veg Burger, Aloo Shorts. Actual bar proportions depend on the Minimum/Maximum settings; Automatic does not always mean a zero baseline.

Show Bar Only

1. Select B2:B7 > Conditional Formatting > Manage Rules.

2. Choose the bar rule > Edit Rule.

3. Check Show Bar Only, click OK and Apply/OK.

The cell displays the bar without its number, but selecting the cell still shows its real number in the Formula Bar. To show values and bars in separate columns, enter C2 = B2, fill through C7, then apply Show Bar Only to C2:C7.

Live change

Changing the Aloo Shorts sales from 10000 to 50000 makes its bar the longest; restoring 10000 returns the earlier order. No reapplication is necessary.

LESSON 30 | Data Bar Customization and Pivot Tables

Fixed threshold via scale Minimum

Example question: visually emphasize sales above 25000. A fixed Data Bar minimum of 25000 starts the scale at that value. This is not a strict Greater Than formatting condition: values at or below the starting point have zero visible positive bar length. The data remains intact.

Apply custom minimum and maximum

1. Select 29_DataBars!B2:B7 and clear previous rules if needed.

2. Home > Conditional Formatting > Data Bars > More Rules.

3. Select Format Style: Data Bar; Minimum Type: Number, Value: 25000.

4. Leave Maximum Automatic, choose Solid Fill, click OK.

Sales	Above 25000	Illustrative bar (maximum 45000)
25000	0	0%

Sales	Above 25000	Illustrative bar (maximum 45000)
30000	5000	25%
35000	10000	50%
45000	20000	100%

Use a fixed target scale

Set Minimum = Number 25000 and Maximum = Number 50000 using Manage Rules > Edit Rule. Then 30000 uses roughly 20%, 35000 uses 40%, and 45000 uses 80% of the available positive bar range. A fixed maximum is helpful when comparing different Slicer selections on the same reference scale.

Correct range and Pivot Table scope

Manually formatting just the currently high-performing cells misses future qualifying cells. Apply the rule to the full field, or in a Pivot Table choose a scope such as All cells showing Sum of Overall Sales values for Product Name. This tends to track changes in product rows better than a fixed cell range. Keep Grand Total out of individual-product comparisons when possible.

Pivot Table procedure

1. Select sales data; Insert > PivotTable; Product Name to Rows, Overall Sales to Values (Sum).

2. Click a product-level sales value > Home > Conditional Formatting > Data Bars > More Rules.

3. Choose the Pivot Table field-aware scope, set Minimum Number = 25000 and Maximum Automatic, click OK.

4. Insert an Order Type Slicer via PivotTable Analyze > Insert Slicer; test Online and Physical Visit.

Important: With an automatic maximum, the bar scale can rescale after Slicer selections; with a fixed maximum, cross-view bar lengths are comparable. If you need an explicit business rule to color cells strictly over 25000, use Highlight Cells Rules > Greater Than instead.

LESSON 31 | Two- and Three-Color Scales

Definition

Color Scales color the whole cell according to its value, rather than adding a horizontal bar. Two-color scales use minimum/maximum color points; three-color scales add a midpoint. Between points, Excel interpolates colors.

Practice and steps

Use the same 10-day daily-sales dataset from Lesson 28 in sheet 31_ColorScales.

1. Select B2:B11.

2. Home > Conditional Formatting > Color Scales.

3. Choose Green-Yellow-Red, with high = green and low = red; inspect the hover tooltip to confirm order.

4. Clear the rule and try Green-White-Red. Copy sales to another column and compare with Blue-White-Red and a two-color preset.

Preset	High	Middle	Low
Green-Yellow-Red	Green	Yellow	Red
Green-White-Red	Green	White	Red
Blue-White-Red	Blue	White	Red
Green-White (2-color)	Green	Interpolated	White

Expected: 10842 receives the high-end color and 4943 receives the low-end color. Intermediate values receive blended shades. Reversing a preset reverses what those colors communicate.

Choose Data Bars or Color Scales

Data Bars emphasize magnitude through length; Color Scales emphasize rank/distribution through whole-cell shading. Both reevaluate as values change.

LESSON 32 | Custom Color Scales

Three reference points

For a three-color scale, configure Minimum (low color), Midpoint (middle color), and Maximum (high color) through Home > Conditional Formatting > Color Scales > More Rules or Manage Rules > Edit Rule.

50th versus 75th percentile

Use the same 10 sales figures. Sort ascending to understand distribution: 4943, 5800, 6200, 6500, 7100, 7800, 8200, 8900, 9300, 10842. The median / 50th percentile is 7450. With Excel PERCENTILE.INC(B2:B11,0.75), the 75th percentile is 8725.

1. Select sales B2:B11 > Conditional Formatting > Color Scales > More Rules.

2. Choose 3-Color Scale; Minimum = Lowest Value (red), Maximum = Highest Value (green).

3. Set Midpoint Type = Percentile, Value = 75, color white; click OK.

Expected: White midpoint shifts from about 7450 under a 50th-percentile rule to about 8725 under the 75th-percentile rule. This does not mean 75% of the maximum value.

Percent vs Percentile vs Number

Type	How to interpret
Percent	Relative distance between minimum and maximum; e.g. 75% from 4000 to 10000 = 8500.
Percentile	Based on distribution; in this dataset the inclusive 75th percentile = 8725.
Number	Fixed numeric reference regardless of distribution changes.

Fixed midpoint 5000

1. Manage Rules > select the 3-color rule > Edit Rule.

2. Set Midpoint Type = Number, Value = 5000.

3. Keep minimum red and maximum green; apply.

Expected: Almost all observed sales values exceed 5000, so most appear on the white-to-green side. For Pivot Table values, prefer the proper field-aware Applies To scope.

LESSON 33 | Icon Sets and Thresholds

Types and purpose

Icon set type	Purpose
Directional arrows	Up / sideways / down performance indication
Shapes and indicators	Colored circles, ticks, warning signs or crosses
Ratings	Stars/bars corresponding to ratings or service levels

Icon Sets show an icon beside each numerical value. Default three-icon thresholds are often percentage-based; customize them when exact targets matter.

Practice dataset: 33_IconSets

Product	Sales
Pizza	12000
Burger	9900
Sandwich	8500
Pasta	15000
Fries	9200

Apply arrows with fixed thresholds

1. Select sales B2:B6 > Home > Conditional Formatting > Icon Sets > 3 Arrows.

2. Conditional Formatting > Manage Rules > select rule > Edit Rule.

3. For both thresholds choose Type Number.

4. Set top threshold 10000; second threshold 9500; click OK.

Sales condition	Icon / status
>= 10000	Green up arrow: target met
>= 9500 and < 10000	Yellow sideways arrow
< 9500	Red down arrow

Expected: Pizza and Pasta green; Burger yellow; Sandwich and Fries red.

CSAT rating thresholds

Example ratings: 8.2, 5.8, 6.3, 7.5, 6.7. Choose a three-icon rating preset; set threshold Types to Number, upper 6.5 and lower 6. Results: 8.2, 7.5, 6.7 = good; 6.3 = medium; 5.8 = poor.

Other icon options

Option	Effect
Reverse Icon Order	Reverses icon assignment direction.
Show Icon Only	Hides displayed number; value still exists.
No Cell Icon	Suppresses the icon for a chosen category.
Threshold Type: Formula	Can reference a target cell when supported by the rule.
Manage Rules	Modify icons, thresholds, Applies To and priority.

One icon for meeting a threshold

To show only a green icon at sales >= 10000, set both thresholds to 10000 with Type Number, and select No Cell Icon for the middle and bottom categories. Use the right icon for the top category.

Pivot Table and Slicer integration

Apply icons to the relevant Pivot Table value field using its field-aware scope. Slicers change displayed results; fixed Number thresholds stay constant. Do not use one mixed threshold rule across revenue, percentage discounts, and customer ratings.

LESSON 34 | Duplicate and Unique Values

Built-in duplicate check

Ticket ID	Customer
T101	Arun
T102	Sneha
T103	Ravi
T101	Priya
T104	Kiran
T105	Meena
T103	Rahul
T106	Anu

1. Create sheet 34_DuplicateValues and enter the table in A1:B9.

2. Select A2:A9.

3. Home > Conditional Formatting > Highlight Cells Rules > Duplicate Values.

4. Choose Duplicate, pick a red format, click OK.

Expected: both occurrences of T101 and both of T103 are highlighted. All occurrences, including the first, are highlighted. Replace the second T101 with T107 and highlights for T101 disappear automatically.

Unique versus duplicate

The Duplicate Values dialog can switch from Duplicate to Unique to highlight values that appear only once in the selected range. Clear earlier rules first if necessary.

Whole-row duplicate limitation

Selecting a multiple-column table and choosing Duplicate Values tests individual cells, not whether entire records match. Repeated customer names or order types may highlight despite rows being different. It does not remove duplicates; use Data > Remove Duplicates for removal after review.

Formula-based exact row matches (additional practical technique)

If columns A:C describe a record, select A2:C9 > Home > Conditional Formatting > New Rule > Use a formula to determine which cells to format. Enter:

Formula: =COUNTIFS($A$2:$A$9,$A2,$B$2:$B$9,$B2,$C$2:$C$9,$C2)>1

Choose a fill. This flags rows whose values match in all three specified columns. Extend references to include other fields when required. Built-in duplicate matching is generally case-insensitive.

LESSON 35 | Top / Bottom and Average Rules

Available commands

Excel command	Effect
Top 10 Items / Bottom 10 Items	Ranks selected values and highlights a chosen count (can change 10 to 3).
Top 10% / Bottom 10%	Highlights top or bottom percentage of observations.
Above Average / Below Average	Compares values to the arithmetic mean of selected numeric data.

Practice data: 35_TopBottomRules

Enter these 12 daily sales values in B2:B13: 6500, 8200, 10842, 7100, 9300, 5800, 4943, 7800, 8900, 6200, 9800, 4500 (dates 01-Nov through 12-Nov in A2:A13).

Top 3 AND Bottom 3 on one range

1. Select B2:B13; clear previous Conditional Formatting for that range.

2. Home > Conditional Formatting > Top/Bottom Rules > Top 10 Items; change 10 to 3; set green fill; OK.

3. Keep B2:B13 selected > Top/Bottom Rules > Bottom 10 Items; change 10 to 3; set red fill; OK.

4. Open Manage Rules to see two rules applying to the same range.

Top three sales (green)	Bottom three sales (red)
10842	4500
9800	4943
9300	5800

Change the 7100 sale to 3000: the Bottom 3 automatically becomes 3000, 4500, 4943; 5800 loses its red fill. Restore 7100.

Top 10 / Bottom 10 and percentages

On 12 values, Top 10 Items highlights all except the bottom two; Bottom 10 Items highlights all except the top two. Top 10% / Bottom 10% selects a fraction rather than a fixed count; ties can mean more than the nominal number get highlighted.

Above / Below Average example

For values 4000, 6000, 8000, 10000, mean = 7000; the last two are Above Average and first two Below Average. Rules can overlap on small or tied datasets; rule priority, ranges, and Stop If True affect the result.

LESSON 36 | Integrated Conditional Formatting Report

Practice dataset: 36_ConditionalAnalysis

Agent	Type	Product	Sales	Orders	CSAT	Discount
Adrian	Online	Pizza	12000	8	8.2	5%
Adrian	Visit	Burger	8500	6	6.2	12%
Albion	Online	Burger	9500	7	5.8	15%
Albion	Visit	Pizza	14000	10	8.5	6%
Roche	Online	Pizza	15000	9	7.8	8%
Roche	Visit	Burger	10500	7	6.8	11%

Correct totals and weighted averages

This six-row dataset contains already summarized groups. Total sales = 69500; total orders = 47. A simple average of group CSATs differs from an order-weighted average. When each row represents a group of orders, calculate weighted CSAT = SUMPRODUCT(CSAT_range, Orders_range) / SUM(Orders_range); similarly for discount. These are approximate summary measures and assume the listed group averages are appropriate to weight by orders.

Calculated from the sample: weighted CSAT = 7.37; order-weighted discount = 9.06%. The course exercise also demonstrated target-based evaluation; always calculate actuals consistently before formatting.

Formatting design by metric

Metric	Appropriate Conditional Formatting	Why
Overall sales / orders / CSAT	Icon Set or formula rule vs target	Shows target achieved / not achieved.
Agent sales	Three-color scale	Separates high, middle, and low.
Agent order counts	Two-color scale	Shows variation without overstating performance judgment.
CSAT	Three-icon ratings, Number thresholds 6 and 6.5	Displays quality status.
Discount	Warning indicator, reverse favorable direction	High discounts may exceed limits.
Product revenue	Three-color scale or Data Bars	Compares product contribution.
Product rating / discount	Separate icon rules with appropriate thresholds	Prevents unit mismatches.

Agent-wise outputs

Agent	Sales	Orders
Adrian	20500	14
Albion	23500	17
Roche	25500	16

Applying a target-referenced rule

Suppose B2 = 69500 Actual Sales and C2 = 70000 Target. Select B2 > Home > Conditional Formatting > New Rule > Use a formula to determine which cells to format. Enter =B2>=C2 and choose green; add a second rule =B2<C2 and choose red. If C2 changes, formatting responds automatically. The transcript also demonstrates using target cell references in eligible Icon Set thresholds (Type = Formula).

Discount percentage threshold setup

For an 8% target, the raw Excel number is 0.08. If higher discount is undesirable, use favorable icons for values <= 0.08 and warnings for larger values. One possible policy: <= 8% good; > 8% through 10% warning; > 10% high warning. Choose icon order/thresholds carefully (reverse icon order where appropriate).

Integrating Pivot Tables and Slicers

1. Create or use compatible Pivot Tables for overall, agent, and product metrics.
2. Format each value field separately with a suitable Conditional Formatting rule and Pivot Table-aware scope.
3. Click a Pivot Table > PivotTable Analyze > Insert Slicer > Order Type.
4. Select the Slicer > Slicer > Report Connections > check compatible Pivot Tables.
5. Test Online and Physical Visit; verify values and icons/scales.
6. For consistent comparisons between Slicer views, use fixed thresholds or scales where appropriate.

Short checklist

Correct data types and totals; use thresholds matching units; distinguish relative versus fixed targets; exclude Grand Totals from individual comparisons if they distort the scale; use Manage Rules for applies-to areas and priority; refresh Pivot Tables when source data changes; use color and icons sparingly for readability.

One-Page Reference | All Lessons

Goal	Excel route / method
Highlight values over a number	Home > Conditional Formatting > Highlight Cells Rules > Greater Than
Solid or gradient bars	Home > Conditional Formatting > Data Bars
Data Bars without visible values	Manage Rules > Edit Data Bar > Show Bar Only
Custom bar minimum / maximum	Data Bars > More Rules > Number or Automatic
Two- or three-color gradient	Home > Conditional Formatting > Color Scales
75th-percentile midpoint	Color Scales > More Rules > 3-Color Scale > Midpoint: Percentile 75
Fixed midpoint 5000	Edit Color Scale > Midpoint: Number 5000
Green/yellow/red performance icons	Icon Sets > More Rules > Number thresholds
Show icons without values	Edit Icon Rule > Show Icon Only
Duplicate or unique values	Highlight Cells Rules > Duplicate Values > Duplicate/Unique
Duplicate full rows	New Rule > Use a formula > COUNTIFS on all columns
Top 3 / Bottom 3	Top/Bottom Rules > Top 10/Bottom 10 Items > change to 3
Top 10% / Above Average	Top/Bottom Rules > corresponding option
Edit or remove formatting	Conditional Formatting > Manage Rules / Clear Rules
Slicer controlling several Pivot Tables	PivotTable Analyze > Insert Slicer; Slicer > Report Connections

Terminology and correctness notes

Bar scales, colors, and icon thresholds are visual classification rules, not a substitute for correct calculations. Relative formats can rescale when filtered. A Data Bar minimum is not equivalent to a strict condition. Built-in Duplicate Values checks individual cells, not whole rows. Weighted averages need the appropriate weights; aggregated averages should not be summed.